

HR Workforce Analytics and Employee Attrition Dashboard

The **HR Workforce Analytics and Employee Attrition Dashboard** project focuses on analyzing employee workforce data to understand salary, performance, job satisfaction, attendance, leave, overtime, training, promotion, and employee attrition. The project uses Microsoft Excel for data cleaning, transformation, VLOOKUP-based data integration, and exploratory analysis, followed by Power BI for interactive dashboard development.

Objective of the project:

The objective of this project is to clean, transform, and analyze employee data to identify key HR metrics and generate meaningful insights about employee performance, satisfaction, attendance, training, promotions, and attrition. The project aims to support better HR decision-making by presenting accurate and useful employee information through data analysis and visualization.

No.	Column	Description
1	Employee ID	Unique identification number assigned to each employee
2	Employee Name	Name of the employee
3	Gender	Gender of the employee
4	Age	Age of the employee in years
5	Department	Department in which the employee works
6	Job Role	Job position or role of the employee
7	Location	Work location of the employee
8	Joining Date	Date on which the employee joined the organization
9	Education	Highest educational qualification of the employee

No.	Column	Description
10	Years of Experience	Total years of professional work experience
11	Manager	Manager or reporting person of the employee
12	Monthly Salary	Employee's monthly salary
13	Performance Rating	Rating 1-5 given to the employee based on performance
14	Job Satisfaction	Employee's level of satisfaction with their job 1-10
15	Attendance Percentage	Percentage of attendance recorded for the employee
16	Leave Days	Number of leave days taken by the employee
17	Leave Type	Type/category of leave taken
18	Overtime	Indicates whether the employee worked overtime
19	Training Hours	Number of hours spent in training
20	Promotion Status	Indicates whether the employee received a promotion
21	Attrition	Indicates whether the employee left the organization

PHASE 1

Data Cleaning and Transformation:

1. **Duplicate Identification**
Conditional Formatting → Highlight Cells Rules → Duplicate Values was used to identify duplicate employee records.
2. **Duplicate Removal**
The Remove Duplicates option was used to remove duplicate records.
 - Initial records: 3,200
 - Duplicate records removed: 200

- Final records: 3,000

3. Missing Value Treatment

- Gender: 48 missing values were replaced using the mode (Male).
- Department: 44 missing values were replaced with Unknown.
- Location: 46 missing values were replaced with Unknown.
- Joining Date: 43 missing values were identified in the Joining Date column. During the data cleaning process, these missing values were initially marked as "Unknown". They were later changed back to blank values to preserve the original missing-data status. In Power BI, a custom column named Joining Date Status was created to classify these records as Available or Missing.
- Education: 47 missing values were replaced with Unknown.
- Manager: 48 missing values were replaced with Unknown.
- Years Of Experience: No missing or Unknown values were found, so no action was required.
- Age: The Age column was checked for missing values and data consistency. The age was calculated using the MEDIAN () function because they have outliers and used to replace the missing Age values.

4. Data Standardization

An inconsistent location value, Bengaluru, was standardized to Bangalore using Find & Replace. A total of 40 replacements were completed.

5. Final Verification

The Employee Master table was checked again after cleaning to ensure that duplicates, missing values, and inconsistent entries had been appropriately handled.

The cleaned Employee Master table was then used for VLOOKUP integration with the HR Metrics data.

HR Metrics Data Cleaning

The HR Metrics data was checked for missing values and incorrect entries. Missing numerical values were replaced using the median, while missing categorical values were replaced with Unknown.

Step 1: Monthly Salary

- Missing values identified: 49
- Replacement method: Mean
- Excel function used: MEAN ()
- The calculated mean value was used to replace the missing salary values.

- Monthly Salary was then formatted in Indian Rupees (₹).

Step 2: Performance Rating

- Missing values identified: 47
- Replacement method: Median
- Excel function used: MEDIAN ()
- The median performance rating was used to replace the missing values.

Step 3: Job Satisfaction

- Missing values identified: 47
- Replacement method: Median
- Excel function used: MEDIAN ()
- The median value was used to replace the missing job satisfaction values.

Step 4: Attendance Percentage

- Missing values identified: 47
- Replacement method: Median
- Excel function used: MEDIAN ()
- The median value was used to replace the missing attendance values because they have outliers.
- Attendance was formatted as a percentage.

Step 5: Leave Days

- Missing values identified: 45
- Replacement method: Median
- Excel function used: MEDIAN ()
- The median value was used to replace the missing leave-day values.

Step 6: Leave Type

- Missing values identified: 46
- Replacement method: Unknown
- The missing Leave Type values were replaced with Unknown because the correct leave category could not be determined.

Step 7: Overtime

- Missing values identified: 45
- Replacement method: Mode ()

- The missing Overtime values were replaced with most frequently occurring values (No) because the actual overtime status could not be determined.

Step 8: Overtime Consistency

An inconsistent value “Y” was identified in the Overtime column.

- Incorrect value: Y
- Correct value: Yes
- Total replacements: 66

The Find & Replace option was used to standardize Y → Yes.

Step 9: Leave Type Consistency

An inconsistent value “sick” was identified in the Leave Type column.

- Incorrect value: sick
- Correct value: Sick
- Total replacements: 647

The values were standardized using Find & Replace.

VLOOKUP Data Integration

After completing the data cleaning process, the Employee Master and HR Metrics tables were integrated using VLOOKUP in Microsoft Excel.

Step 1: Identify the Common Key

The common field between the two tables was Employee ID.

Employee ID was used as the lookup key because it identifies each employee and allows the HR Metrics information to be matched with the corresponding employee.

Step 2: Open the Final HR Data

A separate Final HR Data table was prepared to combine the required employee information and HR metrics.

Step 3: Use VLOOKUP

The VLOOKUP function was used to retrieve the required HR Metrics values based on Employee ID.

The required fields were brought into the Final HR Data, such as:

- Employee Name
- Gender
- Age
- Department
- Job Role

- Location
- Joining Date
- Education
- Years Of Experience
- Manager
- Monthly Salary
- Performance Rating
- Job Satisfaction
- Attendance
- Leave Days
- Leave Type
- Overtime
- Training Hours
- Promotion Status
- Attrition

Step 4: Verify the Results

The Final HR Data was checked to ensure that the values were correctly matched with the corresponding Employee IDs.

Step 5: Prepare the Final Dataset

After verifying the VLOOKUP results, the Final HR Data was used as the main dataset for creating Pivot Tables, Pivot Charts, and further HR analysis.

Pivot Tables and Pivot Charts

The following 17 Pivot Tables and Pivot Charts were created for HR workforce and attrition analysis:

1. Employee Count by Department — Clustered Column Chart
2. Attrition by Department — Stacked Column Chart
3. Average Monthly Salary by Department — Clustered Bar Chart
4. Average Performance Rating by Department — Clustered Column Chart
5. Average Job Satisfaction by Department — Clustered Bar Chart
6. Leave Type Distribution — Doughnut Chart
7. Overtime by Department — Stacked Bar Chart
8. Average Training Hours by Department — Clustered Column Chart

9. Promotion Status by Department — Clustered Column Chart
10. Attrition by Job Role — Stacked Bar Chart
11. Attrition by Gender — Clustered Column Chart
12. Attrition by Job Satisfaction — Line with Markers
13. Attrition by Overtime — Clustered Column Chart
14. Attrition by Performance Rating — Line with Markers
15. Employee Count by Location — Clustered Bar Chart
16. Average Monthly Salary by Job Role — Clustered Bar Chart
17. Employee Count by Department & Gender (*Attrition used as a filter*) — Clustered Column Chart

Key Findings:

The HR data analysis provided the following key observations:

1. **Workforce Distribution:** Employee counts can be compared across departments and locations to understand workforce distribution.
2. **Attrition:** Attrition patterns can be analyzed across departments, job roles, gender, job satisfaction, overtime, and performance ratings.
3. **Salary Analysis:** Average monthly salary can be compared across departments and job roles to understand compensation patterns.
4. **Employee Satisfaction:** Job satisfaction analysis helps identify differences in employee satisfaction levels.
5. **Performance Analysis:** Performance ratings can be compared with employee attrition to understand possible relationships between performance and retention.
6. **Training and Development:** Training hours and promotion status provide insights into employee development and career growth.
7. **Leave and Overtime:** Leave type and overtime analysis helps understand employee work patterns and workload.
8. **Data Quality:** Removing duplicates, treating missing values, and standardizing inconsistent values improved the quality and consistency of the HR dataset.
9. **Dashboard Preparation:** The cleaned and analyzed dataset is ready to be used for the next phase of the project in Power BI.

Conclusion:

The Phase 1 HR data analysis successfully transformed the raw employee dataset into a clean and analysis-ready dataset. Duplicate records were removed, missing values were treated using appropriate methods such as median and Unknown, and inconsistent values were standardized.

VLOOKUP was used to integrate the Employee Master and HR Metrics data using Employee ID as the common key. The final HR dataset was then used to create 17 Pivot Tables and 17 Pivot Charts covering workforce distribution, salary, performance, satisfaction, training, promotion, overtime, leave, and attrition.

Overall, Phase 1 provides a strong foundation for Phase 2 – Power BI, where the cleaned HR data can be used to create an interactive HR Workforce Analytics and Employee Attrition Dashboard.

PHASE 2

Phase 2 of the project is a continuation of Phase 1, where the employee dataset was cleaned, transformed, and prepared using Microsoft Excel. After completing the data cleaning and transformation process, the final dataset containing 3,000 employee records and 21 columns was imported into Power BI for advanced analysis and visualization.

The main objective of Phase 2 was to transform the cleaned HR data into an interactive Power BI dashboard that provides meaningful insights into workforce distribution, employee performance, attendance, leave, training, compensation, promotion, and employee attrition.

The dashboard was designed with interactive filters, drill-down, drill-through, bookmarks, reset functionality, and page navigation to make HR analysis easier and more effective.

Power BI Data Preparation and Transformation

The cleaned dataset from Phase 1 was imported into Power BI. Before creating the dashboard visuals, additional data preparation and transformation were performed to make the dataset suitable for HR analytics.

Joining Date Status – Custom Column

A custom column named Joining Date Status was created because some employee records contained missing Joining Date values.

The column categorizes the records into:

- Available
- Missing

This classification helps identify whether the Joining Date information is available for each employee and makes the missing-date records easier to analyse.

Data Modelling:

The Power BI data model was structured using Employee Master as the central table.

Relationships were created between Employee Master and the HR Metrics table using the common Employee ID field. Similarly, the Final HR Data table was connected to Employee Master

using Employee ID. This centralized data model enabled employee-level information and HR metrics to work together effectively for interactive analysis, DAX calculations, filtering, and dashboard visualizations.

DAX Calculated Columns

To support meaningful HR analysis, the following calculated columns were created using DAX:

- Age Group – categorizes employees into meaningful age groups.
- Experience Level – categorizes employees based on their years of experience.
- Salary Band – groups employees according to salary ranges.
- Performance Category – classifies employees based on performance rating.
- Attendance Category – groups employees according to attendance percentage.
- Satisfaction Category – categorizes employees based on job satisfaction.
- Leave Category – groups employees according to leave-related information.
- Training Category – categorizes employees based on training hours.
- Attrition Status – supports the identification and analysis of employees based on attrition status.

These calculated columns helped convert numerical and categorical HR information into meaningful analytical groups.

DAX Measures

A separate measure table was created to organize the DAX measures used throughout the dashboard.

The following measures were created:

- Total Employees
- Attrition Count
- Attrition Percentage
- Average Age
- Average Monthly Salary
- Average Attendance Percentage
- Average Job Satisfaction
- Average Performance Rating
- Average Years of Experience
- Average Overtime Count
- Average Overtime Rate
- Promotion Count
- Promotion Rate Percentage
- Total Leave Days
- Total Training Hours
- Cumulative Employees

These measures were used across KPI cards, charts, and analytical visuals to provide dynamic results based on selected filters.

Dashboard Development

The Power BI report was developed using five analysis pages and one drill-through Employee Details page.

The dashboard follows a consistent professional design with KPI cards, charts, slicers, navigation controls, and interactive analysis features.

HR Overview

The HR Overview page provides a high-level summary of the organization's workforce.

KPI Cards

The following KPIs were included:

- Total Employees
- Attrition Rate Percentage
- Average Performance Rating
- Average Attendance Percentage

These KPIs provide an immediate overview of workforce size, employee attrition, performance, and attendance.

Employees by Department – Column Chart

Visual: Column Chart

Axis: Department

Values: Total Employees measure

Purpose:

These visual compares employee distribution across different departments and helps identify departments with higher or lower workforce strength.

Employees by Gender – Donut Chart

Visual: Donut Chart

Legend: Gender

Values: Total Employees measure

Purpose:

This visual shows the gender distribution of employees and provides an overview of workforce composition.

Attrition Overview – Donut Chart

Visual: Donut Chart

Legend: Attrition Status

Values: Employee count / Total Employees measure

Purpose:

This visual provides a quick comparison between employees who have remained with the organization and employees who have left.

Employee Joining Trend & Overtime – Combo Chart

Visual: Combo Chart

Purpose:

These visual combines employee joining information with overtime-related analysis to identify workforce and overtime patterns over time.

Employees by Location – Map

Visual: Map

Location: Employee Location

Values: Employee count

Purpose:

The map provides a geographical view of employee distribution across different locations.

Slicers

- Department
- Gender
- Location
- Job Role

These slicers allow users to interactively filter the entire page according to selected workforce characteristics.

Drill-down

The Employees by Department visual supports drill-down from:

Department → Job Role

This allows users to move from department-level workforce analysis to a more detailed job-role level.

Employee & Department Analysis

This page focuses on workforce characteristics, departmental comparison, salary, performance, satisfaction, experience, and training.

KPI Cards

- Total Employees
- Average Age
- Average Years of Experience
- Average Training Hours
- Average Job Satisfaction

Employees by Job Role – Bar Chart

Visual: Bar Chart

Axis: Job Role

Values: Total Employees measure

Purpose:

Shows the distribution of employees across different job roles.

Average Salary by Department – Column Chart

Visual: Column Chart

Axis: Department

Values: Average Monthly Salary measure

Purpose:

Compares average monthly salary across departments and helps identify differences in departmental compensation.

Performance Rating by Department – Combo Chart

Visual: Combo Chart

Axis: Department

Values: Performance Rating measure

Purpose:

Used to compare employee performance levels across departments.

Job Satisfaction by Department – Donut Chart

Visual: Donut Chart

Legend: Department

Values: Average Job Satisfaction measure

Purpose:

Provides a visual comparison of job satisfaction across departments.

Years of Experience vs Average Salary – Scatter Chart

Visual: Scatter Chart

X-axis: Years of Experience

Y-axis: Average Monthly Salary

Purpose:

Shows the relationship between employee experience and salary and helps identify compensation patterns.

Training Hours by Department – Bar Chart

Visual: Bar Chart

Axis: Department

Values: Total / Average Training Hours

Purpose:

Compares training activity across departments and helps identify departments receiving relatively more or less training.

Slicers

- Department
- Gender
- Location
- Job Role

Attendance, Leave & Training Analysis

This page focuses on employee attendance, leave, overtime, training, and promotion-related metrics.

KPI Cards

- Average Leave Days
- Overtime Rate Percentage
- Promotion Rate Percentage

Average Attendance Percentage – Line Chart

Visual: Line Chart

Axis: Time-based analysis

Values: Average Attendance Percentage

Purpose:

Shows attendance trends and helps identify changes in employee attendance patterns.

Leave Type Distribution – Donut Chart

Visual: Donut Chart

Legend: Leave Type

Values: Employee count

Purpose:

Shows the distribution of different leave types used by employees.

Training Hours Analysis – Column Chart

Visual: Column Chart

Values: Training Hours

Purpose:

Provides a visual analysis of employee training activity and helps identify training patterns.

Promotion Status Analysis – Combo Chart

Visual: Combo Chart

Values: Promotion Count / Promotion Rate Percentage

Purpose:

Analyses promotion status and provides an overview of employee career progression.

Overtime Analysis – Donut Chart

Visual: Donut Chart

Values: Overtime information

Purpose:

Shows the distribution of employees according to overtime status and helps identify overtime patterns.

Slicers

- Department
- Gender
- Location
- Leave Type

Attrition & Performance Analysis

This page focuses specifically on employee attrition and its relationship with demographic, performance, satisfaction, salary, and organizational factors.

KPI Cards

- Attrition Employees
- Average Age of Attrition Employees
- Average Salary of Attrition Employees

Attrition by Department – Bar Chart

Visual: Bar Chart

Axis: Department

Values: Attrition Count

Purpose:

Identifies departments with comparatively higher or lower employee attrition.

Attrition by Gender – Donut Chart

Visual: Donut Chart

Legend: Gender

Values: Attrition Count

Purpose:

Compares attrition across gender groups.

Attrition by Location – Tree Map

- Visual: Tree Map
- Group: Location
- Values: Attrition Count
- Purpose: Shows the distribution of employee attrition across different locations and helps identify locations with comparatively higher or lower attrition.

Attrition by Job Role – Column Chart

Visual: Column Chart

Axis: Job Role

Values: Attrition Count

Purpose:

Identifies job roles where employee attrition is comparatively higher or lower.

Performance vs Attrition – Column Chart

Visual: Column Chart

Axis: Performance Category / Rating

Values: Attrition Count

Purpose:

Analyses the relationship between employee performance and attrition.

Performance vs Job Satisfaction – Scatter Chart

Visual: Scatter Chart

X-axis: Job Satisfaction

Y-axis: Performance Rating

Legend: Department

Size: Count of Employees

Purpose:

Examines the relationship between employee satisfaction and performance while allowing comparison across departments.

Slicers

- Department
- Gender
- Location
- Leave Type

Insights & Recommendations

The Insights & Recommendations page summarizes the major findings obtained from the HR analytics dashboard and converts them into actionable recommendations.

Key Insights:

Workforce Distribution

Analysis of workforce distribution across departments, job roles, gender, and locations helps identify the organizational structure and concentration of employees.

High or Low Attrition Areas

Department, location, and job-role analysis helps identify areas where employee attrition is relatively high or low.

Performance Patterns

Performance analysis helps identify patterns in employee performance across departments and understand how performance relates to other workforce factors.

Attendance Trends

Attendance analysis helps identify employee attendance patterns and areas that may require management attention.

Compensation Observation

Salary analysis across departments, experience levels, and employee groups helps identify compensation patterns and differences within the workforce.

Recommendations:

Attrition Reduction Actions

Organizations can focus on departments and job roles showing comparatively high attrition and investigate the underlying reasons for employee turnover.

Employee Engagement Initiatives

Improving employee engagement and job satisfaction through appropriate workplace initiatives can support employee retention.

Training Recommendation

Departments with lower training exposure can be provided with suitable learning and development opportunities to support employee growth and performance.

Promotional & Career Development

Clear career progression and promotion opportunities can encourage employees to remain engaged and support long-term retention.

Attendance Improvement

Attendance trends can be monitored regularly, and appropriate employee-support initiatives can be introduced where attendance concerns are identified.

Employee Details – Drill-through Page

A dedicated Employee Details page was created using Power BI's drill-through functionality.

Drill-through Field

- Month

The drill-through page provides detailed employee-level information, based on the selected month.

This allows users to move from summary-level analysis to a more detailed view of employee information while maintaining the selected month context.

Interactive Dashboard Features

Several interactive Power BI features were implemented to improve dashboard usability.

Slicers

Slicers were added to allow users to dynamically filter dashboard information by relevant employee characteristics such as:

- Department
- Gender
- Location
- Job Role
- Leave Type

Visual-level Filters

Visual-level filters were applied where necessary to remove blank or unknown values from individual visuals and ensure that the charts presented meaningful information.

Drill-down

The HR Overview department visual supports drill-down from:

Department → Job Role

This provides more detailed analysis without creating an additional visual.

Drill-through

The Employee Details page provides employee-level information through drill-through functionality.

Bookmark

A bookmark was created to store the default dashboard/filter state.

Reset Button

A Reset button was placed on the HR Overview page and connected to the bookmark. When selected, it returns the dashboard filters and slicers to the default state.

Page Navigation

Navigation buttons/icons were added to allow users to move easily between the different dashboard pages.

Dashboard Design and Theme

A consistent professional design theme was applied throughout the report.

The dashboard design includes:

- Consistent page layout
- KPI cards for important metrics
- Clearly labelled charts
- Slicers for interactive filtering
- Navigation buttons
- Consistent formatting
- Clean visual hierarchy
- Professional colour theme
- Interactive analysis features

The design was kept simple and user-friendly so that important HR insights can be identified quickly.

Overall Analysis and Key Findings

The Power BI dashboard provides an integrated view of the organization's workforce and employee-related metrics.

The analysis covers:

- Workforce distribution
- Department and job-role composition
- Gender and location distribution
- Employee experience
- Salary and compensation
- Performance
- Job satisfaction
- Attendance
- Leave
- Overtime
- Training
- Promotion

- Employee attrition

The combination of KPI measures, analytical visuals, slicers, drill-down, and drill-through functionality allows users to analyse HR data from both a high-level organizational perspective and a detailed employee-level perspective.

Conclusion

The second phase of the HR Workforce Analytics and Employee Attrition Dashboard successfully transformed the cleaned Phase 1 HR dataset into an interactive Power BI analytics solution.

Using DAX calculated columns, measures, interactive visualizations, slicers, filters, drill-down, drill-through, bookmarks, reset functionality, and page navigation, the dashboard provides a structured approach to understanding workforce characteristics and employee behaviour.

The dashboard helps HR stakeholders identify workforce distribution, performance patterns, attendance and training trends, compensation differences, promotion patterns, and potential attrition areas. The Insights & Recommendations page further converts the analytical findings into practical HR actions.

Overall, the project demonstrates how cleaned HR data can be transformed into a professional Business Intelligence solution that supports data-driven workforce analysis, employee retention, performance management, and HR decision-making.